



Master data

Course 2 - Module 2 - Lesson 2.2.2

Master data: configuration, contextual and reference datasets

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I. Recap use cases and selected use case

Use cases

1. Dynamic population estimates and migration
2. Tourism
3. Transportation and commuting

Use case: Transportation and commuting

Mobility patterns are spreading and the transport authority of a metropolitan area needs to update the city transport system to meet current demand.



Use case: Transportation and commuting

Mobility surveys are expensive and are limited to a small sample, and it is decided to obtain the origin/destination matrix from MPD.

Results of main origin/destination pairs by time of the day and weekday (working vs. weekend days) will be compared with public transport current supply.

Use case: Transportation and commuting

Real world example: Origin - Destination (OD) matrices in Salvador, Brazil.

Analysis of travel demand in Salvador municipality (2.4M inhabitants) for the local government

- Study population: residents in Brazil
- Trips under study: origin or destination in Salvador metropolitan area
- Days of study: 15 days of 2024, covering weekdays, weekends and the *Carnival* period
- Zoning system: 385 zones
- Products: trip matrices
- Segmentation by time of the day, trip purpose, trip distance, sociodemographic profile

Use case: Transportation and commuting

Total trips production (residents and nonresidents), Salvador (Brazil) Metropolitan Region)



Use case: Transportation and commuting

Explore more transportation use cases at:

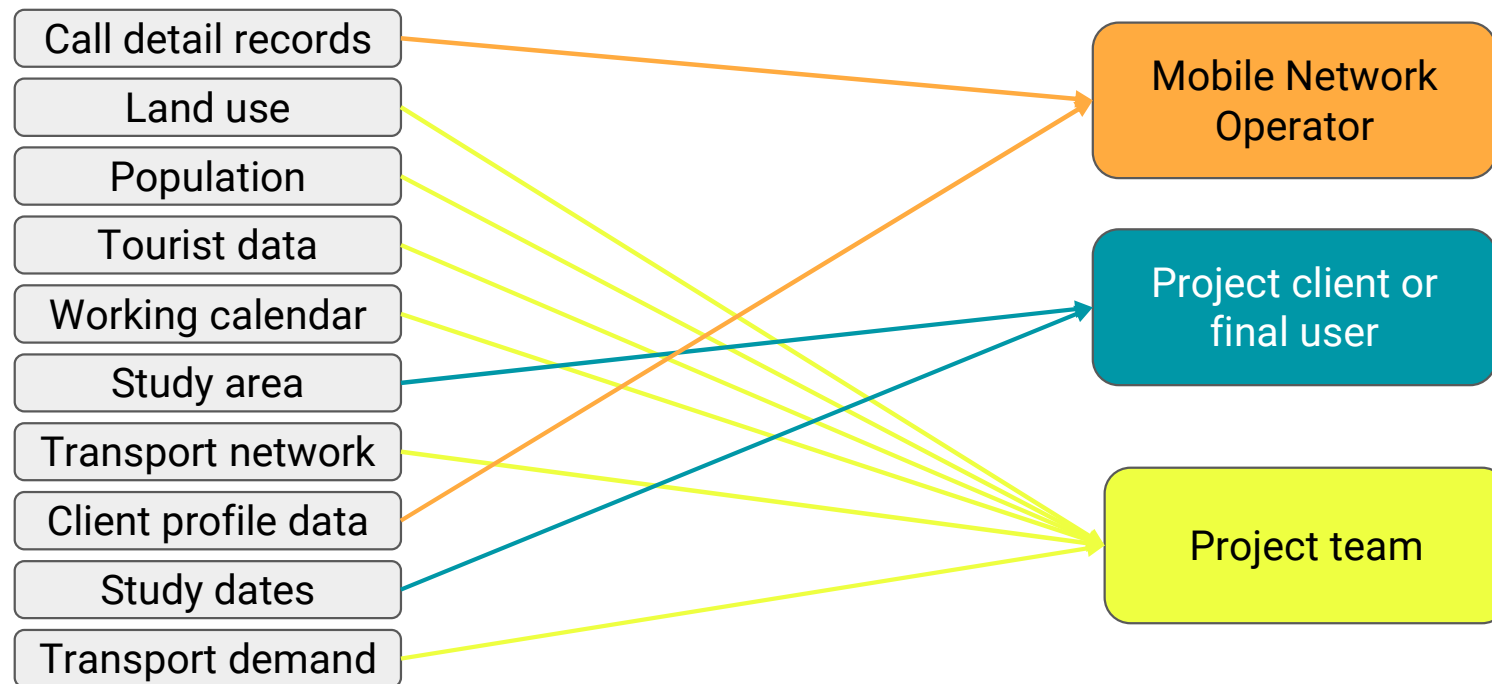
[4. Applications of Mobile Phone Data in Transportation - Methodological guide on the use of mobile phone data: Transport and Commuting - UN Statistics Wiki](#)

- 4.1 Transport Statistics
 - Case Study: Replacing Indonesia's Household Travel Surveys
- 4.2 Transport and Urban Planning
 - Case Study: UK's Use of MPD to Understand Modal Usage
 - Case Study: Italy's use of MPDs to estimate the O_D matrix
- 4.3 Transit & Public Transport Planning
 - Case Study: Tracking Rail Passengers in Austrian Railways
 - Case Study: Seoul's Use of MPD in Planning Night Bus Service
 - Case Study: Dubai RTA's Bus Station Layout Optimization Using MPD
 - Case Study: Italy's Use of MPD in Train Management
 - Case Study: Commuters in Tartu, Estonia
- 4.4 Traffic Analysis
 - Case Study: Poland's Use of MPD in Traffic Analysis
- 4.5 Operational Decision Support
 - Case Study: Dubai's Use of MPD in Events Management
- 4.6 Events and Crises Management
 - Case Study: Dubai's use of MPD in understanding pedestrian movements in EXPO 2020
- 4.7 Traffic & Transit Demand Prediction
 - Case Study: Italy's use of MPD to Study Traffic Impact

2. Configuration, contextual and reference data

What data do you need and who will typically provide it?

Online activity



Configuration, contextual and reference data

Configuration data are datasets which are needed for specifying the scope of the Mobile Phone Data (MPD) analysis. For example: list of dates to analyse, study zoning in which to produce the final indicators, etc.

Contextual data comprises auxiliary datasets such as administrative data, roads and buildings, which support MPD processing to improve accuracy and provide contextual enhancements.

Reference data then helps validate and assess the quality and representativeness of MPD. Sociodemographic data, for instance, can be used to expand on each user's trips or activities in order to represent the behaviour of other residents with similar characteristics.

3. Configuration data needs and examples

Configuration data

Configuration data defines the **project scope** and needs to be agreed with the final client to achieve the best trade-off between cost, delivery time, user/trip characterization and privacy impact.

- Which is the **target population**?
 - Residents of a specific study zone
 - All residents of a given country
 - Residents and visitors
 - Tourists
 - Excursionist
 - Tourists and excursionists

Configuration data

- Which is the **study area**?
 - The city centre
 - The metropolitan area, how is it spatially defined?
 - The whole region or country?
- Which are the **trips** under study?
 - Trips with origin and destination within the study area
 - Trips with origin and/or destination within the study area
 - Trips which route intersects the study area

Configuration data

- Which is the **zoning system**?
 - Administrative units, like districts or census tracts
 - *Ad hoc* units, like transport zones
- * Is it the same for origin/destination and for place of residence?
- * To be reviewed by the project team to ensure Mobile Phone Data (MPD) **spatial resolution**

Configuration data

- Which is the **study period**?
 - A selection of days within different:
 - Years, to monitor mobility
 - Seasons (winter, summer, rainy, dry, ...) to measure different mobility patterns across the year
 - Special events (sport or music events, Carnival, Easter holidays, weekends,...) to analyse the impact of particular events.
 - * To be reviewed by the project team to ensure MPD availability on selected days

Configuration data

- Which is the **temporal resolution**?
 - Within the study period
 - Each day individually
 - Average working day
 - Period within the day:
 - Daily
 - Typical peaks and valleys: morning peak, morning valley, mid-day peak, afternoon valley, afternoon peak, night
 - By hour of the day
 - * Typically period when the trips starts, could be when the trip ends.

Configuration data

- Which are the **sociodemographic variables of interest**?
 - Age, in age groups
 - 5-year intervals
 - Life-cycle intervals (0-14, 15-24, 25-44, 45-64, 65-74, 75 and over)
 - ...
 - Sex, typically male, female as in most statistics
 - Income
 - Average net income per household
 - Average net income per person
 - ...

Configuration data

- Think of other variables the final user might be interested in.

Configuration data

- Example of other variables the final user might be interested in:
 - ☐ Distance
 - ☐ Trip duration
 - ☐ Trip purpose
 - ☐ Transport mode
 - ☐ User place of residence
 - ☐ Professional vs. personal mobility
 - ☐ Trip legs and route (*home - train station - airport - airport - hotel*)

Configuration data

Example

Requirement topic	Description	Notes
Study population	Residents in Brazil and foreign visitors	
Study area	Salvador metropolitan area	
Study trips	Trips with origin or destination in the study area	
Zoning system	• Origin/destination matrix: Transport zones • Place of residence: ○ Study area: census tract ○ Rest of the country: municipality ○ Rest of the world: country	Files with spatial definitions (geojson, geopackage, shp,...) and unique ID. Transport zones provided by the client. To be compared with cell-tower distribution. Administrative units: latest available from official statistics.

Configuration data

Example

Requirement topic	Description	Notes
Temporal resolution (study period)	- Average working day (Tuesday to Thursday) - Average weekend day (Saturday and Sunday) 10 Carnival days	Exact dates to be agreed between the client and the data provider
Temporal resolution (within the day)	24 hours	Period is referred to the hour when the trip starts
Age	Life-cycle intervals: 0-19, 20-44, 45-64, 65-74, 75 and over	
Sex	Male, female	
Place of residence	- Origin purpose and destination purpose - Categories: - Home - Work - Other frequent location - Occasional location	

Configuration data

Example

Requirement topic	Description	Notes
Distance intervals	• 0-2] Km • (2-5] Km • (5-10] Km • (10-20] Km • (20-50] Km • > 50 Km	Distance can be measured as a straight line from origin and destination (less time consuming) or through the transport network (time and cost consuming)
Transport mode	• Active mobility (walking, bicycle, other personal vehicles) • Collective public transport (bus, metro, tram) • Private motor vehicle (motorcycle, car, taxi)	Transport survey to be provided by the client
Trip purpose	• Study area: census tract • Rest of the country: department • Rest of the world: country	
Mobility type	Only personal mobility	Trips made by professional drivers while they are working will be excluded.

Configuration data

Review activity

- Write a detailed requirement list for the case study exposed on slides 5 and 6
- Add as many questions for clients as needed
- Or add as many details as needed to the study case

4. Contextual and reference data topics and data sources

Contextual and reference data

Contextual data allows us to determine the type of activity a user might be engaged in during the day.

Reference data helps validate and assess the quality and representativeness of Mobile Phone Data.

Multiple data sources can be used either as contextual or reference data in different project steps.

Contextual and reference data

Main data topics

- Sociodemographics and other statistical datasets:
 - Population by place of residence, age, sex, nationality, income, ...
 - Tourism: visitors statistics (tourists and excursionists)
- Activities location: land use, Points of Interest (PoI)
- Transport system:
 - Transport supply (roads, terminals, schedules,...)
 - Transport demand and mobility surveys (also called transport surveys)
- Spatial boundaries: administrative and other vector boundaries (land use, international borders) vs. grids
- Other auxiliary data:
 - Codes and relationships: Mobile Country Code (MCC) and (Mobile Network Code) MNC lists and country codes
 - Holidays and working days
 - Weather
 - ...

Contextual and reference data

The following slides offer a review of the main datasets required.

For each topic, you will find:

- Why and on which step it needs to be used
- Quality criteria to improve results
- Where to find it:
 - Indication whether data is typically covered by national institutes
 - At least one global data source if available.

Contextual and reference data | Sociodemographics & other statistical datasets

- **Population** by place of residence, age and sex
 - Registered resident population $\neq$ actual population volume and distribution
 - Still, it is needed to scale results to the resident population
 - ◎ Population scaling is covered on [2.4.4 Aggregates to Indicators](#) lesson
 - If the Mobile Network Operator (MNO) provides age and sex details for their clients, population scaling benefits from introducing these variables
 - Estimating user's place of residence requires a longitudinal analysis.
 - ◎ Home location process is covered on [2.4.2 Meaningful Locations and UE](#) lesson

Contextual and reference data | Sociodemographics & other statistical datasets

- **Population** by place of residence, age and sex
 - Quality improves if these criteria are maximized
 - Spatial resolution
 - Reference date closest to the project study period
 - Small age groups
 - Population values by age and sex

Contextual and reference data | Sociodemographics & other statistical datasets

- **Population** by place of residence, age and sex
 - Typically obtained from national statistics institutes and available at least every 10 years (census)
 - Additional data to estimate population on a more recent date if needed
 - Official estimates or population projections,
 - Or combining:
 - Number of births every year
 - Number of deaths by age or age group
 - Migration balance
 - WorldPop provides population estimates by sex and age in a 100 meter grid.

WorldPop

<https://hub.worldpop.org/project/categories?id=8>

Contextual and reference data | Sociodemographics & other statistical datasets

- **Tourism:** visitor statistics
 - Foreign visitors can be divided into:
 - Tourists: if they spent at least one night
 - Excursionists: if they arrive and depart from the country in the same day
 - This data is needed to scale roaming in users
 - ◎ [Population scaling](#) is covered on [2.4.4 Aggregates to Indicators](#) lesson

Contextual and reference data | Sociodemographics & other statistical datasets

- **Tourism:** visitor statistics
 - Quality improves if these criteria are maximized
 - Temporal resolution:
 - Monthly data: short holidays like Carnival or Easter can be considered
 - Quarterly data: summer vs winter seasons can be distinguished
 - Yearly data: preferably post-covid
 - Reference date closest to the project study period
 - Territory of origin: country based will be easier to match with SIM card nationality if needed

Contextual and reference data | Sociodemographics & other statistical datasets

- **Tourism:** visitor statistics

- Typically obtained from national statistics institutes from tourism datasets
- Because figures are obtained for tourism purposes, the most common variables are:
 - Number of overnights at hotels, hostels and other paid accommodations
 - Number of arrivals at hotels, hostels and other paid accommodations
 - Sometimes there is no data available for excursionists
 - Some countries, like Spain, have specific border statistics. See FRONTUR survey results [here](#).
 - Useful variable: average number of overnights by day and country of origin
- UN tourism provides total arrivals by destination country and visitor type (tourist, excursionist) or region of origin, and total yearly up to 2022. [145 key tourism statistics](#).



Contextual and reference data | Activity location

- Land use, Points of interest (Pol)
 - Land use and/or Pol are needed to identify users meaningful locations and usual environment
 - Land use: all territory is covered and a trade-off between spatial and attribute resolution and feasibility leads to aggregates on both dimensions that works well with MPD spatial resolution
 - Pol: can be used to refine or validate particular locations. For example, overlaying urban land use and education Pol to indicate whether work related trips or presence could be study trips instead.
 - Satellite imagery can be helpful for detailed checks, or even to generate land use maps.
 - Work location process is covered on [2.4.2 Meaningful Locations and UE](#) lesson
 - Plausibility checks are covered on [2.5.1 Aggregate and Indicator Quality Assessment](#) lesson

Contextual and reference data | Activity location

- Land use, Points of interest (Pol)
 - Quality improves if these criteria are maximized
 - Spatial resolution and Minimum Mapping Unit (MMU), particularly in urban studies
 - Reference date closest to the project study period
 - Land use classification: residential vs. commercial, industrial and other work-related categories is preferred over a single “built-up land” category
 - Pol can be used to further classify land use classes

Contextual and reference data | Activity location

- Land use, Points of interest (Pol)

- Land use maps are typically obtained from national statistics and/or geographic institutes, updated every 5-10 years.



WorldCover V2 2021
WorldCover-Map

The ESA provides a World Land cover map for 2021 covering all countries. There is a single built-up area category. <https://esa-worldcover.org/en/data-access#citation>.



Global Land
Analysis & Discovery

- GLAD's Global land cover and land use 2019 shows the percentage of built-up area on each grid cell. <https://glad.umd.edu/dataset/global-land-cover-land-use-v1>.



Copernicus LAC provides a land use layer of 2023 for Latin America and the Caribbean, also with a single artificial land category. <https://cobertura-suelo.copernicuslac-chile.eu/en>



- Pol can be obtained from [Open Street Maps](#), for example with the [Overpass turbo API](#) .

overpass turbo

Contextual and reference data | Transport system

- **Transport supply**

- Transport supply includes:
 - Road network
 - Public transport terminals and stops: train, bus, tram,
 - Passenger ports
 - Commercial airports
 - Schedules
- Transport supply data is needed to generate results by transport mode.

○ This is not covered on this course edition

Contextual and reference data | Transport system

- **Transport supply**
 - Quality improves if these criteria are maximized
 - Reference data close or previous to the study date. We do not want new transport infrastructures that could impact route or mode choice.
 - Smart selection of passenger or cargo terminal locations

Contextual and reference data | Transport system

- **Transport supply**

- Transport supply network is often provided by national geographic institutes.
- The transport supply geographical location can be obtained from [Open Street Maps](#), for example with the [Overpass turbo API](#) .  
- [GTFS](#) is the standard for transit schedules and stops locations. They can be published by the corresponding public transport authority.
- General Transit Feed Specification offers a list of GTFS data providers:



<https://gtfs.org/resources/data/>

Contextual and reference data | Transport system

- **Transport demand**

- Public transport records: each time a passenger accesses a public transport service, a record known as validation is produced. Depending on the mode of transport the validation can be a) *At entry / at exit / at both times* of the service, b) in vehicle (buses) or at station (metro/rail/BRT/cable)
 - If the access is with a transport card, the validation also contains a card identifier that can be used to trace the different accesses to the service of the same passenger. If the access is with a single ticket, this data may not be available.
- Mobility surveys (also called transport surveys): in-depth surveys with detailed data that can be used as input in a machine learning model to estimate transport mode according to user and trips characteristics.

Contextual and reference data | Transport system

- **Transport demand**

- Vehicle counting: this is the number (and type) of vehicles that cross a particular infrastructure (typically road) point.
- Transport demand data and/or mobility surveys can be used to generate results by transport mode and/or to validate origin/destination matrices results.
- This is not covered on this course edition

Contextual and reference data | Transport system

- **Transport demand**
 - Quality improves if these criteria are maximized
 - For public transport, data on entry and exits from the system are preferred. Many public transport systems do not require users to validate when exiting the system, so there are algorithms to estimate the exit stop.
 - For mobility surveys, it is important to consider the date of study. They are typically made every 10 years, and mobility behaviours have changed notably pre vs. post covid-19.
 - For vehicle counts, total number of vehicle by type and day (or even hour of the day).

Contextual and reference data | Transport system

- **Transport demand**

- **Public transport validations** can be provided upon request by the corresponding public transport authority.
- **Mobility surveys** main results are usually available in public reports. Machine learning models need microdata as input, which can be provided upon request by the corresponding statistician institute or transport authority.

Contextual and reference data | Transport system

- Transport demand

- For **vehicle counts**, total number of vehicle by type and day (or even hour of the day) are typically provided by clients interested in knowing the origin and destination of toll users.
- Traffic density metrics (like Average Daily Traffic (ADT) or Annual Average Daily Traffic (AADT)) can be found as open data in some countries.



<https://mapadetrafico.transportes.gob.es/>



 TomTom's Traffic Density API is available in a broad selection of countries:

<https://developer.tomtom.com/traffic-stats/documentation/product-information/market-coverage>

<https://developer.tomtom.com/traffic-stats/documentation/api/traffic-density> .

Contextual and reference data | Spatial boundaries

- Vector geometries vs. spatial grids (raster)

Aspect	Vector	Raster (Grid)
Structure	Points, lines, polygons	Grid of uniform cells (pixels)
Resolution	Variable (based on feature geometry)	Fixed (e.g., 100m, 1km per cell)
Data Types	Discrete (e.g., parcels, tracts)	Continuous or categorical (e.g., land use)
Precision	High geometric accuracy	Limited by cell size
Overlay	Computationally expensive (geometry operations)	Fast (cell-by-cell, matrix-based)
File Size	Often larger (due to complex geometry, attributes)	Often smaller (especially for simple rasters)
Best For	Boundaries, legal zones, infrastructure	Environmental variables, remote sensing
Examples	Census polygons, land parcels	Land cover grids, population rasters

Contextual and reference data | Other auxiliary data

- **Codes and relationships: MCC and MNC lists and country codes**
 - Useful when tackling roamers or detecting national users abroad
 - Mobile Country Codes (MCC) identify the country to which a mobile subscriber belongs.
 - By combining both the MCC and Mobile Network Codes (MNC) you get the HNI (Home Network Identity). This is essentially a single string of the MCC and MNC combined, known in international roaming contexts as the PLMN (Public Land Mobile Network).
 - MCC, MNC and ISO country names and codes: <https://mcc-mnc.net/>
 - Full ISO-3166 country names and codes (Alpha-2, Alpha-3, Numeric): <https://www.iso.org/obp/ui/#search/code/>

Contextual and reference data | Other auxiliary data

- **Holidays and working days**

- Consider the correct working days in your country (i.e. Monday to Friday, Monday to Saturday,...)
 - Consider at least national holidays, or any event that could alter the selection of the home location
 - Also useful in plausibility checks: you might (must) see different patterns on national holidays.
-
- ◎ Home location process is covered on [2.4.2 Meaningful Locations and UE](#) lesson
 - ◎ Plausibility checks are covered on [2.5.1 Aggregate and Indicator Quality Assessment](#) lesson

Contextual and reference data | Other auxiliary data

- **Holidays and working days**



National and local holidays and observances, including clock change dates, weekly updated.

<https://www.timeanddate.com/holidays/>



National and other holidays calendar Python library.

[holidays·PyPI](#)

- It is recommended to check official calendars in the study area.

Contextual and reference data | Other auxiliary data

- **Weather**

- Try to avoid dates with extreme weather like rain or high temperature in the home location selection process.
- Use historical weather data to check your results: for example, lower presence in parks on rainy days.
- ◎ Home location process is covered on [2.4.2 Meaningful Locations and UE](#) lesson
- ◎ Plausibility checks are covered on [2.5.1 Aggregate and Indicator Quality Assessment](#) lesson

Contextual and reference data | Other auxiliary data

- **Weather**

- National weather agencies typically provide current and sometimes historical data.
- Meteostat provides worldwide weather data, including hourly data at the station level.

 Meteostat Developers <https://dev.meteostat.net/>

5. An example of how to validate the selected use case results with reference data

How to validate the selected use case results with reference data

Use case

- Mobility patterns are spreading and the transport authority of a metropolitan area needs to update the city transport system to meet current demand.
- Mobility surveys are expensive and are limited to a small sample, and it is decided to obtain the origin/destination matrix from MPD.
- Results of main origin/destination pairs by time of the day and weekday (working vs. weekend days) will be compared with public transport current supply.

How to validate the selected use case results with reference data

Use case results example (basic od matrix)

dates <i>string</i> <i>yyyymmdd-yyyymmdd</i>	day_type <i>string</i>	hour <i>string</i>	origin <i>string</i>	destination <i>string</i>	trips <i>numeric</i>
20250204-20250206	working_day	06-07	0101001	0101001	532
20250204-20250206	working_day	07-08	0101001	0101002	1258
20250207-20250209	weekend	07-08	0101002	0102002	2598
...	...	...	...	...	...

How to validate the selected use case results with reference data

Use case results example (od matrix with optional fields)

dates <i>string</i> yyyymmdd-yyyymmdd	day_type <i>string</i>	hour <i>string</i>	origin <i>string</i>	destination <i>string</i>	age_group <i>string</i>	sex <i>string</i>	trip_purpose <i>string</i>	trip_distance <i>string</i>	user_residence <i>string</i>	...	trips <i>numeric</i>
20250204- 20250206	working_day	06-07	0101001	0101001	25-44	female	work	500-1000	0101001	...	10
20250204- 20250206	working_day	06-07	0101001	0101001	25-44	male	work	500-1000	0101001	...	100
20250204- 20250206	working_day	06-07	0101001	0101002	25-44	male	work	1000-5000	0101001	...	359
...	...	...	...	...	...	...	...	...	...	...	...

How to validate the selected use case results with reference data

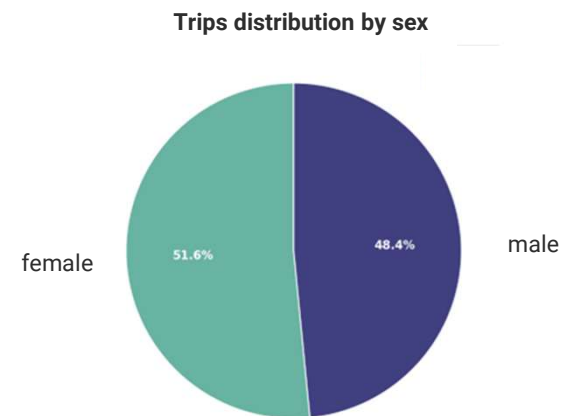
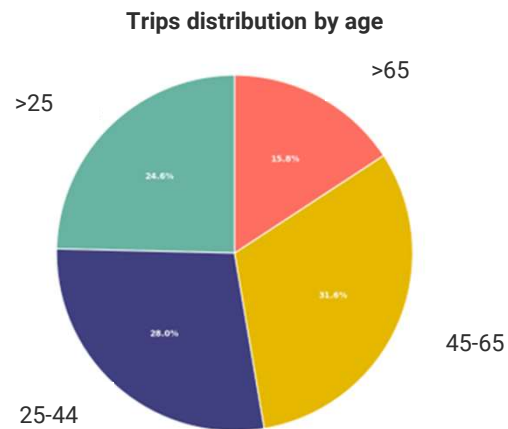
The following content is just a brief example on a few datasets that can be used to validate results.

- © Plausibility checks are covered on [2.5.1 Aggregate and Indicator Quality Assessment](#) lesson

How to validate the selected use case results with reference data

Mobility (or transport) survey

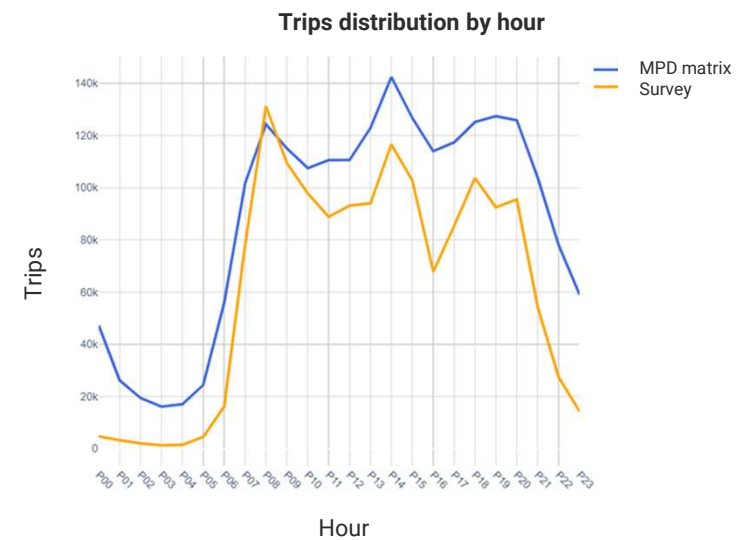
- Comparing general distribution by user profile: age, sex, distance travelled...



How to validate the selected use case results with reference data

Mobility (or transport) survey & weather conditions

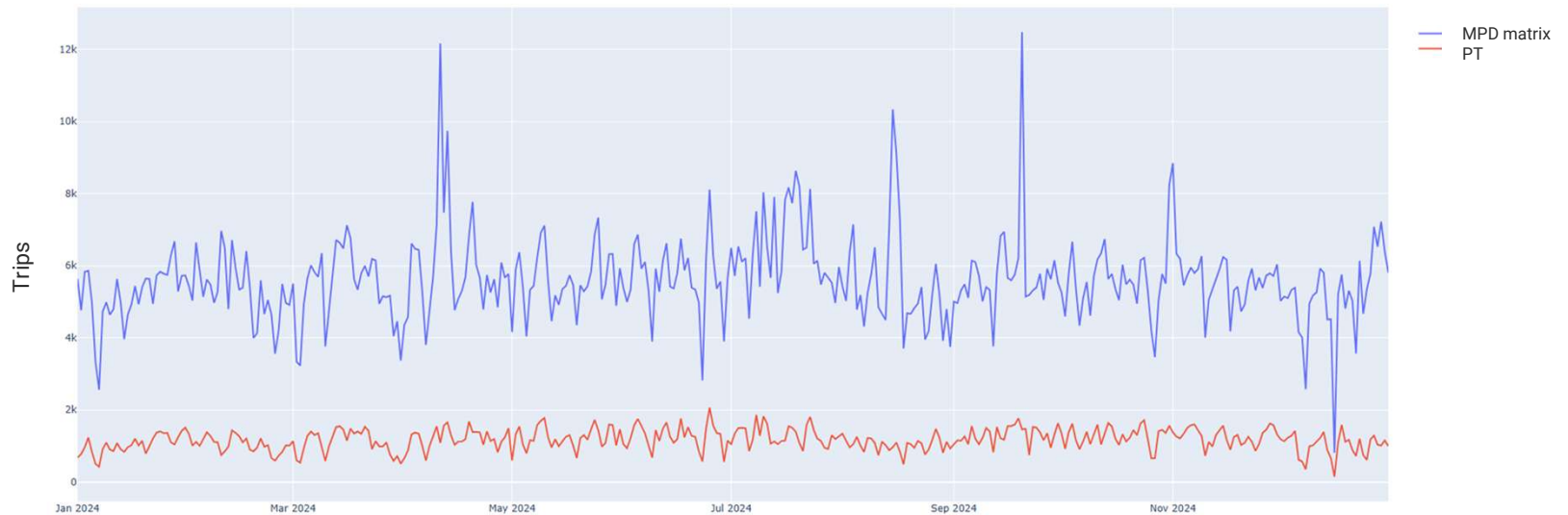
- Comparing working days peaks and valleys. Should they remain similar to the survey? Are changes expected due to:
 - The time gap between the study period and the survey dates?
 - Weather conditions?



How to validate the selected use case results with reference data

Public transport validations

- Overall mobility must be greater than public transport mobility.



How to validate the selected use case results with reference data

Land use

- Are origin/destination purposes related to land use and PoI?
- For example, when trip purpose is Education, is there any educational facility in the destination zone?

6. Key considerations and data summary

Key considerations and data summary

Key considerations

- In addition to MPD and other MNO data, getting useful information from MPD requires additional datasets.
- This lesson covers some of the more usual datasets required.
- There are more attributes that could be of interest, like users income or socioeconomic status, nationality,... Also the type of device to remove IoT or other not personal devices.
- It is important to plan and assign resources to gather and treat the required datasets at the beginning of the project.
- Knowing how data was created and its limitations facilitates proper data usage and correct results interpretation.
- Data maintenance costs must also be considered.
- Most datasets can be used as configuration, contextual and/or reference data. The following slide contains a summary table.

Key considerations and data summary

Topic	Configuration	Contextual	Reference
Population	To decide output trips segmentations	Scaling MPD sample	To compute Trip Generation Rate (TGR) and to validate with external sources
Tourism	To decide whether roamers can be included. To choose nationality groups	Scaling roamers	To compare results, to analyse TGR correctly in highly touristic areas
Pol, land use		Home and work location	To validate trip purposes
Transport supply	To choose modal split categories	Modal split	To validate Public Transport trips origin, destination and time
Transport demand	To choose modal split categories	Modal split	To validate trip volumes
Transport survey	To choose modal split categories	Modal split	To validate modal split distribution, and other attributes present in the survey
MCC and MNC	To choose roamers. To choose nationality groups		
Holidays, weather and other events	To choose study days wisely	Home and work location	To understand unexpected behaviours

Optional activity

Optional activity

Reflect on how to combine the contextual datasets described with those provided by the MNO, such as CDR or cell towers, and the specific purpose for which this would be done.



The GDF-MPD window is a multi-partner window funded by the Spanish Government, Ministry of Economy, Commerce and Business. The GDF-MPD program is jointly managed by DEC (DECDG, DIME), Poverty and Digital Development Global Practices in technical partnership with the International Telecommunications Union (ITU).





Mobile Phone Data for Policy Programme (WB-GDF)
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